



BRAC UNIVERSITY
Department of Computer Science and Engineering
M.Sc. in Computer Science and Engineering Program
Quiz 1, Spring 2026

Course: CSE710 (Advanced Artificial Intelligence), Section-1
Instructor: Dr. Md. Golam Rabiul Alam
Full Marks: 15
Time: 120 minutes

Note: Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. **Determine** the HMM parameters considering the following COVID 19 variants dataset. Afterward, draw the HMM model. [CO3, C4, Marks: 5+10]

States={D: Delta, O: Omicron}					
Observations (Genome Sequence)={M: RBM Spike, B: RDB Spike}					
TimeStamp-->	Sample1	Sample2	Sample3	Sample4	Sample5
State Sequence-1	D	D	D	D	O
Observation Sequence-1	B	B	M	M	M
State Sequence-2	D	O	O	D	O
Observation Sequence-2	M	B	M	B	B
State Sequence-3	D	O	O	D	O
Observation Sequence-3	B	M	M	B	M
State Sequence-4	O	O	D	D	D
Observation Sequence-4	M	M	B	B	B

Now, consider the HMM model as the initial values of HMM learning parameter.
Apply Baum-Welch algorithm (Only one iteration) to fine tune the HMM learning parameters considering the following new observation sequence:
 MBB